Math 73235 Sample Midterm Exam #2 Spring 2025

1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

   (a) The main equation from Gauss' **Divergence Theorem** for a vector field $\vec{F}$ that relates a triple integral over a 3-dimensional bounded region (i.e., volume) T with a surface integral over its boundary surface S with an outward facing unit normal $\hat{n}$.
   Solution. □

   (b) The area element dA for the surface $\phi = \frac{\pi}{4}$ in spherical coordinates.
   Solution. □

   (c) A normal vector for the sphere $x^2 + y^2 + z^2 - 4 = 0$.
   Solution. □

   (d) The curl of ∇f, i.e., $\nabla \times \nabla f$, where $f(r, \theta, z) = \mathrm{e}^r \sin(\theta) - z$. *Hint:* Think before you leap!
   Solution. □

   (e) The expression for the volume integral $\iiint_T f(x, y, z)\, dxdydz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w).
   Solution. □

2. (10 points) Use Gauss' **Divergence Theorem** to find the surface integeral $\oiint_S \vec{F} \cdot \hat{n}\, dA$ for the vector field

$$\vec{F} = x\hat{i} + y\hat{j} + z\hat{k},$$

where S is the surface of a sphere of radius 2.

Solution.

□

3. (10 points) Evaluate $\iiint_T (x + y + z)\ dxdydz$ where T is the volume of the first octant inside the cone $x^2 + y^2 \le 4z^2$, $z \in [0, 2]$, as shown below.

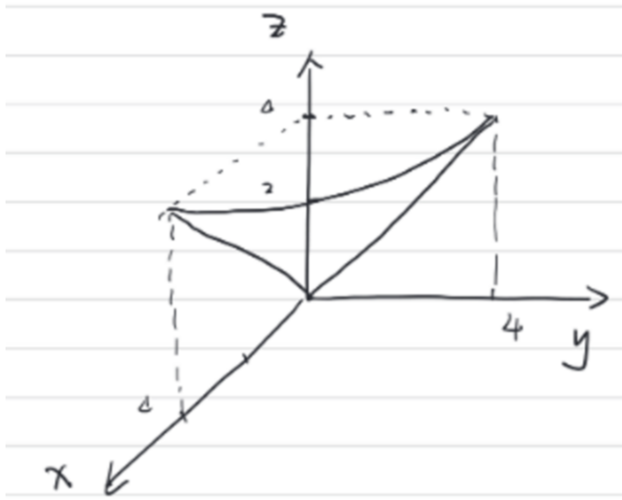

Figure 1: Region T.

Solution.

$\square$

4. (10 points) Verify Stokes' Theorem by computing the following line integral:

$$\oint_C \vec{F} \cdot d\vec{r},$$

where $\vec{F} = -5y\hat{i} + 4x\hat{j} + z\hat{k}$, and C is the circle $x^2 + y^2 = 16$, $z = 4$, traversed counter-clockwise, and then comparing this result to that of the surface integral

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dA,$$

where S is the surface $x^2 + y^2 \leq 4$, $z = 4$.

Solution.

□

5. (10 points) In the Cartesian plane with coordinates (x, y), we may define another set of coordinates(u, v) using the transformation functions

$$x = u^2 + v^2, \quad y = u^2 - v^2.$$

Find both the Lamé coefficients h_u and h_v and thus provide the expression for the line element ds^2 and the area element dA in this new coordinate system.

Solution.

$\square$

Formula Sheet

General Coordinates:

$(x_1, x_2, x_3) \to (q_1, q_2, q_3)$:

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

Derivative Matrix: $D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$

Lamé coefficients: $h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j}\right)^2$

Jacobian: $\frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, 1_3)} = |J| = |\det D| = |h_1 h_2 h_3|$

Line element: $ds^2 = h_1^2 dq_1^2 + h_2^2 d_2^2 + h_3^2 dq_3^2$

Volume element: $dV = h_1 h_2 h_3 \; dq_1 dq_2 dq_3$

Area element for surface with $q_j = const$ $(i, j, k$ distinct): $dA = h_i h_k \; dq_i dq_k$

Gradient for scalar function f: $\nabla f = \frac{1}{h_1}\frac{\partial f}{\partial q_1}\hat{e}_{q_1} + \frac{1}{h_2}\frac{\partial f}{\partial q_2}\hat{e}_{q_2} + \frac{1}{h_3}\frac{\partial f}{\partial q_3}\hat{e}_{q_3}$

Divergence and curl of vector field $\vec{F} = F_{q_1}\hat{e}_{q_1} + F_{q_2}\hat{e}_{q_2} + F_{q_3}\hat{e}_{q_3}$:

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}(h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2}(h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3}(h_1 h_2 F_{q_3})\right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\hat{e}_{q_1} & h_2\hat{e}_{q_2} & h_3\hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f:

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_3 h_1}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right].$$

Cylindrical Coordinates:

$$x = r\cos theta, \quad y = r\sin\theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho\cos\theta\sin\phi, \quad y = \rho\sin\theta\sin\phi, \quad z = \rho\cos\phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho\sin\phi, \quad h_3 = h_\phi = \rho.$$